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Memory-enhanced reinforcement learning energy management framework for coordinated torque allocation in distributed electric vehicles

Huimin Liu, Cheng Lin, Xiao Yu, Zhenyi Tao, Xingyu Xia, Jiaqi Xu, Liqing Zhu, Jinhua Wen

***Abstract*—Torque coordination is critical for energy management in four-wheel distributed electric vehicles capable of independent front–rear torque regulation, yet conventional independent drive and braking strategies neglect the continuity of longitudinal torque regulation. Although unified control frameworks alleviate this issue, the resulting high-dimensional state space and nonlinear actuator coupling make real-time decision-making difficult under long-horizon driving conditions. To address this challenge, this study proposes a unified drive–brake coordinated energy management framework that integrates driving and regenerative braking into a single longitudinal force optimization problem. The torque allocation policy is learned using a memory-enhanced LSTM-TD3 algorithm that captures long-term dependencies through sequence-based state representations. The input sequence horizon is analyzed to identify an effective range for stable training, enabling coordinated front–rear torque allocation in long-horizon complex driving scenarios. Evaluations of real vehicle–road multimodal driving cycles demonstrate improved regenerative braking recovery and motor efficiency while maintaining longitudinal dynamics and torque smoothness. Compared with non-coordinated control and memoryless TD3, the proposed strategy reduces energy consumption by 8.13% and 1.3%, respectively, and deviates by only 5.67% from the theoretical energy lower bound obtained by dynamic programming.**

***Index Terms*—Distributed electric vehicle; Energy management; Drive-brake coordinated control; Memory-enhanced reinforcement learning;**

I. Introduction

With the advancement of vehicle electrification, various powertrain configurations have been proposed. Among them, distributed electric drive systems enable independent wheel-end torque control [1], decouple vehicle dynamics, and offer superior potential for intelligent energy management compared to centralized powertrains [2]. However, the multi-actuator nature of distributed configurations and increasingly complex driving conditions increases the dimensionality of perception and control variables [3], leading to challenges in information processing, actuator coordination, and control latency. These issues hinder online energy management strategy (EMS) optimization and degrade control accuracy [4].

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Huimin Liu, Cheng Lin, Zhenyi Tao, Xingyu Xia, Jiaqi Xu, Liqing Zhu, Jinhua Wen are with the National Engineering Research Center of Electric Vehicles, Beijing Institute of Technology, Beijing 100081, China.

Xiao Yu is with the School of Transportation and Vehicle Engineering, Shandong University of Technology, Zibo 255000, China.

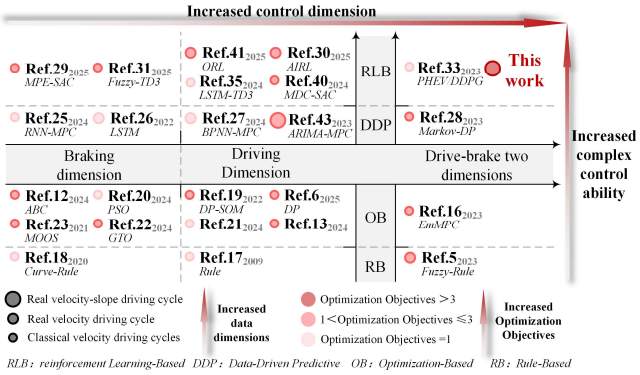


Fig. 1. The literature summary of EMS and the method.

Existing studies on distributed electric vehicles typically treat driving and braking as independent processes [5], neglecting the continuous bidirectional torque regulation capability of wheel-end actuators. As a result, non-coordinated control strategies persist, causing unsmooth torque transitions and limited regenerative energy recovery. Moreover, both optimization-based [6] and data-driven [7] longitudinal control methods struggle to maintain optimality and robustness under high-dimensional perceptual and control spaces, often suffering delays and performance degradation [8].

A. Literature Review

Distributed electric drive configurations employ independently controllable wheel-end motors, offering greater flexibility and optimization potential for traction and regenerative braking energy management than centralized powertrains [9]. This structure significantly increases the degrees of freedom in energy allocation, resulting in strong coupling among vehicle states, energy demand, motor efficiency, and system constraints. This coupling enlarges the state space and decision complexity, making real-time optimal energy regulation difficult for traditional rule-based or model-based strategies under diverse driving conditions [10].

The present study categorizes the current relevant research based on two dimensions: energy management control objects and control methods, as shown in Fig. 1. The following sections will provide a detailed review of the research progress in this field along this framework.

Most existing EMS for distributed configurations inherits the centralized powertrain control paradigm, treating driving and braking as independent processes [11]. Consequently, research

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typically focuses on mode-specific optimization, such as regenerative–mechanical braking coordination [12] or multi-motor torque distribution during driving [13], [14]. From an energy management perspective, this separation restricts the ability of distributed architectures to deliver continuous bidirectional wheel-end torque, leading to unsmooth transitions, suboptimal energy recovery, and unnecessary dissipation of kinetic energy, especially under dynamic conditions such as frequent acceleration–deceleration cycles and varying road gradients [15]. A unified longitudinal energy flow formulation that treats driving and regenerative braking as a single continuous control variable is therefore required to enhance overall energy utilization efficiency [16].

While unified energy management improves efficiency, it also increases real-time decision-making complexity due to high-dimensional state inputs and multi-actuator coupling. Rule-based approaches lack adaptability [17], [18], whereas optimization-based methods, including DP and MPC [19], [20], [21], suffer from excessive computational burden and model dependence [22], [23], limiting real-time applicability under complex conditions [24].

To improve the capability of feature extraction under complex driving conditions, recent studies have introduced data-driven deep learning methods to learn high-dimensional representations of vehicle operating states [25]. However, most of these data-driven models are used only for perception or state estimation tasks [26], while the decision-making layer, particularly for energy allocation, remains dominated by optimization-based methods such as MPC [27]. As a result, the inherent conflict between optimization accuracy and real-time feasibility in such methods persists [28]. Actually, the essence of this contradiction lies in the real-time and coordination problems arising from the strong nonlinear coupling between high-dimensional state inputs and multi-actuator control outputs.

To address the challenge of establishing nonlinear mappings between high-dimensional vehicle states and multiple energy regulation outputs, recent studies have introduced Deep Reinforcement Learning (DRL) into the energy management domain of electric vehicles [29]. Unlike traditional physical-model-based or purely data-driven approaches, DRL can learn end-to-end control policies that directly map complex system states to energy allocation decisions [30]. This capability enables DRL to effectively handle system nonlinearities, high-dimensional perception inputs, multi-actuator coupling, and delayed multi-objective optimization [31], making it particularly suitable for real-time coordinated energy management in distributed electric drive systems [32].

Building on this capability, several studies have applied the Deep Deterministic Policy Gradient (DDPG) algorithm to driving and regenerative braking energy distribution problems in multi-motor architectures, achieving notable improvements in energy efficiency while significantly reducing the reliance on handcrafted control rules [33]. However, DDPG is known to suffer from overestimation bias in value-function learning, which may lead to unstable policy convergence in complex control tasks. The Twin Delayed Deep Deterministic Policy Gradient (TD3) algorithm effectively mitigates this issue through the use of dual critic networks, delayed policy updates, and target policy smoothing, thereby improving training stability and control robustness. Consequently, TD3 has attracted increasing attention in recent studies on energy management and torque coordination for electric vehicles [34].

Furthermore, to better capture the temporal correlations inherent in vehicle dynamics and energy demand, recent research has explored the integration of recurrent neural networks, particularly Long Short-Term Memory (LSTM) structures, into DRL frameworks [35]. By introducing memory mechanisms into the policy network, these approaches enable reinforcement learning agents to leverage historical state information, improving decision-making in long-term control problems. Memory-enhanced DRL architectures, such as LSTM–TD3, have shown improved stability and adaptability in complex control tasks, including active suspension control [36], UAV maneuvering under uncertainty [37], dynamic scheduling in manufacturing [38], and thermal management for battery electric vehicles [39]. However, the application of temporal-memory-enhanced DRL for unified drive–brake energy management in distributed electric vehicles remains underexplored, especially regarding the insufficient research on the time horizon of state inputs under complex, long-duration driving conditions.

Moreover, existing DRL studies mostly rely on simplified velocity profiles [40], [41], neglecting real-world road information such as slope [42], which critically affects power demand and policy generalization [43].

Accordingly, as illustrated in Fig. 1, current research gaps can be summarized as:

- The absence of a unified drive–brake EMS;
- Insufficient integration of real-world vehicle–road data in DRL training;
- A lack of temporal memory mechanisms in the DRL-based energy management strategy.

B. Contribution

This study develops a unified energy management framework for distributed electric vehicles capable of independent front–rear torque regulation to address regenerative energy loss caused by non-coordinated drive–brake control and to improve temporal adaptation under complex vehicle–road conditions. First, because non-coordinated driving and braking may cause regenerative energy loss and torque discontinuity, driving and regenerative braking torques are unified into a single bidirectional longitudinal control variable, leading to a unified EMS framework. Second, because this unified framework introduces stronger nonlinear coupling, higher-dimensional state dependence, and more pronounced temporal interactions under complex vehicle–road conditions, an LSTM-TD3-based strategy is developed for real-time torque allocation. Third, because the effectiveness of memory-enhanced DRL in long-horizon scenarios depends not only on the existence of temporal memory but also on whether the memory horizon matches the vehicle longitudinal response and operating-condition evolution, the input sequence length is further systematically analyzed and optimized. Based on this progressive design, the main contributions of this study are summarized as follows:

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- **Unified drive-brake EMS framework:** driving, regenerative braking, and mechanical braking are coordinated within a single bidirectional front–rear longitudinal control framework.
- **LSTM-TD3 strategy under complex vehicle-road conditions:** A memory-enhanced DRL architecture is developed to handle the strong coupling and high-dimensional temporal dependence introduced by unified drive–brake coordination, with real-world multimodal vehicle–road data incorporated into training.
- **Sequence horizon optimization for memory-based EMS:** The temporal input length is systematically optimized to improve task-relevant temporal information extraction, training stability, and long-horizon energy management performance.

The remainder of this paper is organized as follows. Section II presents the system modeling. Section III formulates the unified longitudinal energy management framework. Section IV introduces the proposed memory-enhanced DRL-based drive–brake coordination strategy. Section V reports the results of the HIL test, and Section VI concludes the paper.

II. System Modelling

A. Configuration of 4WID-EMB Vehicle

This study adopts a four-wheel independent drive and electro-mechanical braking (4WID-EMB) configuration, as shown in Fig. 2. Each wheel is driven by a permanent-magnet synchronous motor (PMSM) through a planetary reduction stage and equipped with an independent EMB unit. Therefore, the proposed architecture is not a direct in-wheel drive configuration, but rather a "motor + planetary reducer + wheel" structure. The main vehicle and component parameters used in this study are summarized in TABLE I, including the vehicle parameters, battery-pack specifications, motor ratings, inverter limits, and EMB actuator characteristics. These parameters are used not only for vehicle modeling but also for enforcing the hardware-related constraints in the proposed control framework.

B. Modeling of 4WID-EMB

1 Electric Drive System Model

The electric drive system is modeled using a quasi-static formulation. The motor efficiency map and torque constraints are both defined in the motor speed–torque domain. Wheel-side torque and power are subsequently derived through gear-ratio and transmission-efficiency transformation, thereby maintaining physical consistency in speed scaling, torque conversion, and efficiency evaluation as described in Eq. (1). Specifically, for each motor, the operating efficiency is obtained by looking up the efficiency MAP (Fig. 3) with the real-time motor speed and torque using linear interpolation; the instantaneous electrical power is then calculated from the motor torque, speed, and the corresponding efficiency, and the energy consumption is obtained by integrating the electrical power over time. The battery-side power is obtained by accumulating the electrical powers of the four motors with the reducer and inverter efficiencies listed in TABLE I consistently included.

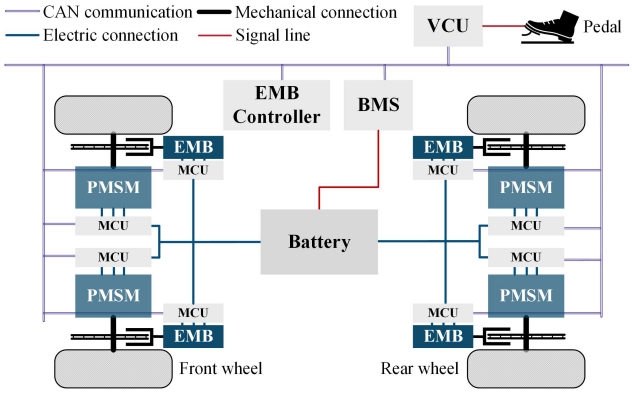


Fig. 2. 4WID-EMB system configurations.

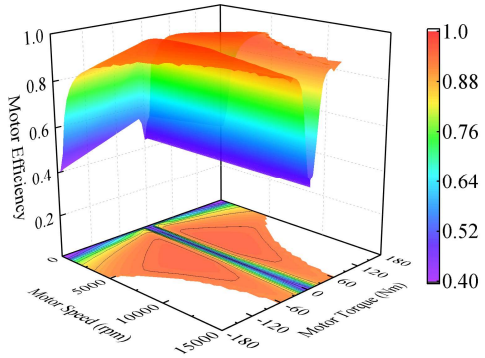


Fig. 3. Motor drive and braking efficiency MAP.

$$\begin{cases} \omega_{M_i} = i_p \cdot \omega_{wi} \\ T_V = \sum_{i=1}^{4} T_{w_i} = \sum_{i=1}^{4} i_p \cdot \eta_p \cdot T_{M_i} \\ P_{Vd} = \sum_{i=1}^{4} P_{M_{id}} = \sum_{i=1}^{4} T_{M_i} \cdot \omega_{M_i} / \eta_{M_i} \cdot \eta_p \\ \eta_{M_i} = f_{motor}(T_{M_i}, \omega_{M_i}) \end{cases} \tag{1}$$

Where P_{Mid}, T_{Mi}, $\omega_{Mi,}$ and η_{Mi} represent the power, torque, speed, and efficiency of PMSM. f_{motor} represents the mapping function of the efficiency, and the MAP of η_{Mi} is shown in Fig. 3. η_p is the transmission efficiency of the planetary gear reducer, T_V and P_{Vd} represent the total torque and power of the electric drive system, and i_p is the planetary gear reducer ratio.

2 Electro-Mechanical Brake System Model

The braking system comprises motors and EMB units, where regenerative braking recovers kinetic energy, and EMBs provide fully decoupled mechanical braking. EMB braking supplies the remaining demand beyond regenerative capability, and the braking force is defined in Eq. (2).

$$\begin{cases} F_\mu = F_{\mu f} + F_{\mu r} = \dfrac{T_{\mu f}}{r} + \dfrac{T_{\mu r}}{r} \\ T_{\mu f} = \sum_{i=1}^{2} i_p \cdot T_{M_i} + \sum_{i=1}^{2} T_{EMB_i} \\ T_{\mu r} = \sum_{i=3}^{4} i_p \cdot T_{M_i} + \sum_{i=3}^{4} T_{EMB_i} \end{cases} \tag{2}$$

where F_μ represents the vehicle braking force, $F_{\mu f}$, $F_{\mu r}$, $T_{\mu f,}$ and $T_{\mu r}$ represent the front and rear axle braking forces and torques, and T_{EMBi} represents the mechanical braking torque output from the EMB system, $i \in [1,2,3,4]$.

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During regenerative braking, the vehicle's kinetic energy is converted into battery electrical energy via the motor and planetary gear reducer. This process is represented in Eq. (3). Where $P_{M_{ib}}$ represents the single motor input power during braking, the braking efficiency MAP is shown in Fig. 3, and P_{bat} is the power produced by regenerative braking of the vehicle.

$$\begin{cases} P_{M_{ib}} = T_{M_i} \cdot \omega_{M_i} \cdot \eta_{M_i} \cdot \eta_p \\ \eta_{M_i} = f_{motor}(T_{M_i}, \omega_{M_i}) \\ P_{bat} = -\sum_{i=1}^{4} P_{M_{ib}} \end{cases} \tag{3}$$

TABLE I
SYSTEM PARAMETERS

System	Parameters	Symbol	Values
Vehicle	Vehicle mass	M	2430kg
	Frontal area	A	2.34m^2
	Coefficient of air resistance	C_d	0.24
	Tyre radius	r	0.35m
	Rolling resistance coefficient	f	0.014
	Wheelbase	L	2760mm
	Vehicle center of mass height	h_g	500mm
	Distance from CG to front axle	a	1320mm
	Distance from CG to rear axle	b	1440mm
Planetary gear	Reducer ratio	i_p	10.6
	Reducer efficiency	η_p	0.985
Battery pack	Cell chemistry	/	NCM
	Voltage	V_{oc}	403.2V
	Capacity	Q_{bat}	240Ah
	Energy capacity	E_{bat}	96.8kWh
	Battery pack mass	M_{bat}	610kg
Motor	Rated power/ Peak Power (per motor)	P_{MR}/P_{MP}	45kW /80kW
	Rated torque/ Peak Torque (per motor)	T_{MR}/T_{MP}	75Nm /180Nm
	Maximum Speed	n_M	15000rpm
	Peak power duration	t_{TMP}	30s
	Efficiency map	/	Shown in Fig. 3
Inverter	DC bus voltage	V_{inv}	403.2V
	Rated current/max current (per inverter)	C_R/C_M	120/220A
	Efficiency	η_I	97%
EMB	Maximum braking torque (per wheel)	T_{EMB_max}	1800Nm per wheel
	Response time	t_{EMB}	150ms
	Rate limit	R_{EMB}	1.3×10^4 Nm/s per wheel

3 Battery System Model

The battery is modeled using an equivalent internal-resistance model, in which the terminal voltage, battery power, and SOC evolution are calculated according to Eq. (4). As shown in Fig. 4 (a), the open-circuit voltage and internal resistance are parameterized as functions of SOC: the former increases nonlinearly with SOC, whereas the latter is relatively high at low SOC, then decreases rapidly and gradually stabilizes in the medium-to-high SOC range. These characteristics indicate that the battery voltage response and power capability are strongly SOC-dependent, and thus must be explicitly captured in the electrical model.

$$\begin{cases} V_{bat} = V_{oc} - I_{bat} R_{bat} \\ P_{bat} = \sum_{i=1}^{4} P_{M_i} = V_{oc} I_{bat} - I_{bat}^2 R_{bat} \\ \Delta SOC = -\dfrac{V_{oc} - \sqrt{V_{oc}^2 - 4R_{bat} \cdot P_{bat}}}{2R_{bat} \cdot Q_{bat}} \Delta t \end{cases} \tag{4}$$

Where V_{bat} is the output voltage, R_{bat} is the equivalent internal resistance, and P_{bat} is the power of the battery.

Since regenerative braking requires the battery to absorb braking energy in addition to supplying driving power, the maximum charging rate constraint as a function of SOC and temperature is considered, as summarized in Eq. (5).

$$C = f_{Charge}(T, SOC) \tag{5}$$

Where f_{Charge} denotes the calibrated mapping of the allowable maximum charging rate in the SOC–temperature domain. Fig. 4 (b) provides its corresponding map, from which it can be observed that the admissible charging rate becomes more restrictive under low-temperature and high-SOC conditions.

4 Vehicle Longitudinal Dynamics Model

The vehicle driving and braking force demands are calculated using the vehicle longitudinal dynamics model.

$$F_t = Mgfcos\theta + \frac{1}{2}\rho C_d A V^2 + Mgsin\theta + M\frac{dV}{dt} = \frac{T_V}{r} \tag{6}$$

Where F_t is the vehicle longitudinal force demand, F_f is the rolling resistance, ρ is the air density (kg/m³), θ is the road gradient, F_w is the air resistance, and F_j is the acceleration resistance.

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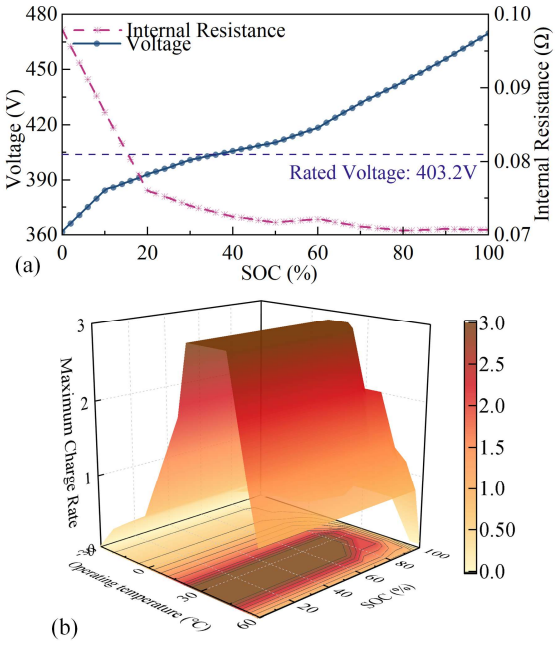


Fig. 4. Battery parameters: (a) SOC-dependent open-circuit voltage and internal resistance; (b) Maximum charge rate.

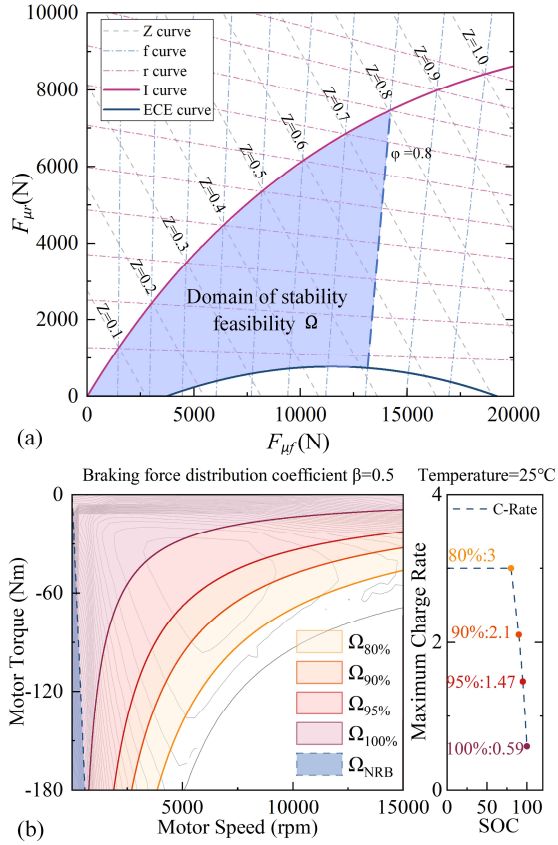


Fig. 5. Braking strategy constraints: (a) feasible front–rear braking torque distribution; (b) regenerative braking boundary.

III. Coordinated Control Framework Design

This section studies longitudinal drive–brake coordinated EMS for distributed electric vehicles capable of independent front–rear torque regulation. It should be noted that the validated controller in this study addresses front–rear longitudinal torque allocation only; left–right torque vectoring, yaw-stability control, emergency braking, and full vehicle-level safety integration are not validated in this work. Braking torque constraints of the 4WID-EMB configuration and regenerative limits imposed by battery charging constraints are first analyzed. The problem is then formulated as a Markov decision process with unified state, action, and reward definitions, enabling a coordinated drive–brake control framework for unified longitudinal torque distribution.

A. Constraints on Axle Brake Force and Regenerative Braking

To ensure braking stability, the front–rear braking forces must lie within the feasible domain defined by the ideal braking force distribution (I-curve) and the ECE regulation, as illustrated in Fig. 5(a) [33]. The shaded area denotes the stability-feasible braking-force region, within which the front and rear axle braking forces satisfy both the adhesion-utilization requirement and the regulatory braking-distribution constraint. The I-curve and ECE boundaries are given by Eq. (7) and (8).

$$\begin{cases} F_{\mu f} = \varphi \cdot \dfrac{Mg}{L}(b + \varphi h_g) \\ F_{\mu r} = \dfrac{1}{2}\left[\dfrac{Mg}{h_g} \sqrt{b^2 + \dfrac{4h_g L}{Mg} F_{\mu f}} - \left(\dfrac{Mgb}{h_g} + 2F_{\mu f}\right) \right] \end{cases} \tag{7}$$

$$\begin{cases} F_{\mu f} + F_{\mu r} = Mg \cdot z \\ F_{\mu f} = \dfrac{Mg(z + 0.07)(b + z h_g)}{0.85L} \end{cases} \tag{8}$$

where φ denotes the tire–road adhesion coefficient, which is equal to the braking strength z under ideal braking-force distribution.

Regenerative braking torque is constrained by the battery's maximum charging rate, with motor braking capability varying with battery temperature and SOC. At low speeds, reduced back-EMF limits regenerative current and recoverable power, yielding the upper bound of regenerative braking torque in Eq. (9) [18]. The resulting regenerative braking constraints are illustrated in Fig. 5(b). The different colored feasible regions in the figure represent the constraints imposed by the maximum allowable charging rate on motor torque under SOC conditions ranging from 100% to 80%. As SOC increases, the feasible range for motor torque decreases, meaning the available regenerative braking torque diminishes. This range prevents overcharging of the battery during regenerative braking.

$$T_{M_i}^{max,b} = -\frac{p \cdot \psi_f}{R_a} \cdot \omega \tag{9}$$

where $T_{M_i}^{max,b}$ is the maximum braking torque at low speeds, p is the number of motor pole pairs, ψ_f is the permanent magnet flux linkage, and R_a is the stator phase resistance.

B. Control Problem Formulation

This study addresses continuous longitudinal drive–brake torque allocation in a 4WID-EMB system. Owing to complex conditions, high-dimensional states, and nonlinear dynamics, conventional methods are ineffective, and the problem is

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therefore formulated as a Markov decision process (MDP) to enable DRL-based near-optimal torque allocation.

The MDP is defined as (S, A, P, R, γ), where S and A denote the state and action spaces, R is the reward function, and $\gamma \in (0,1]$ is the discount factor. In DRL, the environment represents the state transition process, so the transition probability P need not be explicitly modeled.

1) **State Space**

The state vector determines the agent's perception of the vehicle–road environment and is designed to satisfy observability, reward relevance, and normalization. The selected states capture longitudinal motion and driver intent, road-induced energy demand, and battery and powertrain constraints on torque allocation, as defined in Eq. (10).

$$S = \left[V_{real}, V_{tgt}, \delta_{acc}, \delta_{dec}, a, T_d, \theta, P_V, SOC, T_f^T, T_r^T, J_l\right] \tag{10}$$

where V_{real} and V_{tgt} are the real and target vehicle velocity, a is acceleration, δ_{acc} and δ_{dec} are acceleration and braking pedal signals, T_d is the demand torque, θ is the road slope, P_V is the real-time power of the vehicle, and J_l is the longitudinal jerk.

To reduce the influence of different data scales, the state variables are normalized to the range [−1,1] as follows:

$$\tilde{s}_i = 2 \cdot \frac{s_i - s_i^{min}}{s_i^{max} - s_i^{min}} - 1 \tag{11}$$

The normalization bounds of the state variables are set as follows: V_{real} and $V_{\text{tgt}} \in [0, 120]$ km/h; δ_{acc} and $\delta_{\text{dec}} \in [0, 1]$; $a \in [-3, 3]$ m/s²; $T_{\text{d}} \in [-720, 720]$ Nm; $\theta \in [-8\%, 8\%]$; $P_{\text{V}} \in [-150, 320]$ kW; SOC $\in [0, 1]$; T_{f} and $T_{\text{r}} \in [-360, 360]$ Nm (motor-side axle torque); and $J_{\text{l}} \in [-5, 5]$ m/s³.

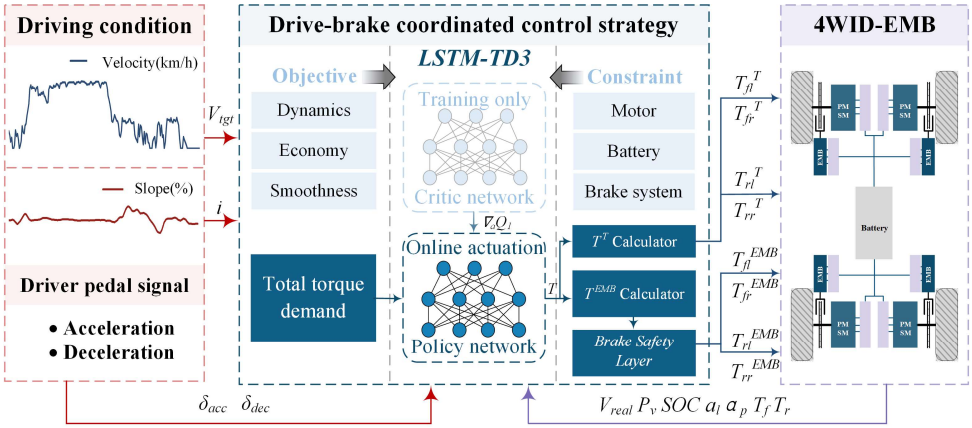


Fig. 6. DRL-based coordinated energy management framework for distributed configuration.

2) **Action Space**

Since longitudinal energy management is governed by front and rear axle torques, only these torques are selected as control variables, and the action space A is defined in Eq. (12).

$$\begin{cases} A = [T_f^T, T_r^T] \\ T_f^T = \sum_{i=1}^{2} T_{Mi} \\ T_r^T = \sum_{i=3}^{4} T_{Mi} \end{cases} \tag{12}$$

It should be clarified that the two action variables in Eq. (12) are defined on the motor side, each representing the summed torque command of the two motors on the corresponding axle, with positive values denoting driving torque and negative values denoting braking torque. Since each axle is equipped with two identical motors with a peak torque of 180 Nm (TABLE I), the motor-side summed axle torque range is [-360, 360] Nm, consistent with the action range listed in TABLE II. The corresponding wheel-side torques are obtained from the motor-side commands through the planetary reduction stage, as formulated in Eq. (1). During braking, when the demanded braking torque does not exceed the maximum available regenerative braking torque provided by the four motors, which is bounded by the motor capability, the battery charging-power limit defined in Eq. (5), and the low-speed boundary defined in Eq. (9), the braking demand is fully supplied by regenerative motor braking; otherwise, the EMB units supplement the remaining mechanical braking torque, as defined in Eq. (2).

3) **Reward Function**

The reward function guides agent learning and affects convergence and performance. To achieve this, dynamics, energy efficiency, and torque smoothness are selected as optimization objectives, and the reward is defined as:

$$\begin{cases} R(t) = -[w_1 R_d(t) + w_2 R_e(t) + w_3 R_j(t)] \\ R_d(t) = \left| V_{tgt}(t) - V(t) \right| \\ R_e(t) = SOC(t-1) - SOC(t) \\ R_j(t) = d^2 V(t) / dt^2 \end{cases} \tag{13}$$

The total reward is the negative weighted sum of the tracking error R_d, energy consumption R_e, and longitudinal jerk R_j, with w_1-w_3 denoting the corresponding weights, where a larger reward indicates better performance. Specifically, the components of the reward function in this research are not equally weighted. Among them, the energy consumption term serves as the core optimization objective, guiding the strategy to prioritize improving the vehicle's overall fuel efficiency, the tracking error term and longitudinal impact term primarily function as auxiliary terms, ensuring that the vehicle maintains basic dynamic response capabilities while suppressing abrupt changes in control outputs, thereby preventing the agent from sacrificing fundamental driving performance in exchange for superficial energy-saving gains. Before weighting, each reward component is mapped to the interval [0, 1] through min–max normalization, so that no term unintentionally

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dominates the policy update due to scale differences. The weight coefficients were then determined through comprehensive comparisons across multiple simulation runs, and the sensitivity of the strategy performance to the selected weights is further analyzed in Section V through a perturbation experiment (TABLE III). The results indicate that the weight combination (w_1=0.3, w_2=0.6, w_3=0.1) most effectively emphasizes the fuel economy optimization objective while ensuring basic driving performance and ride comfort requirements. In addition, actuator and state constraints are enforced in the control framework/environment to prevent infeasible actions before execution. The same constraint-handling logic is applied to all compared methods to ensure a fair comparison.

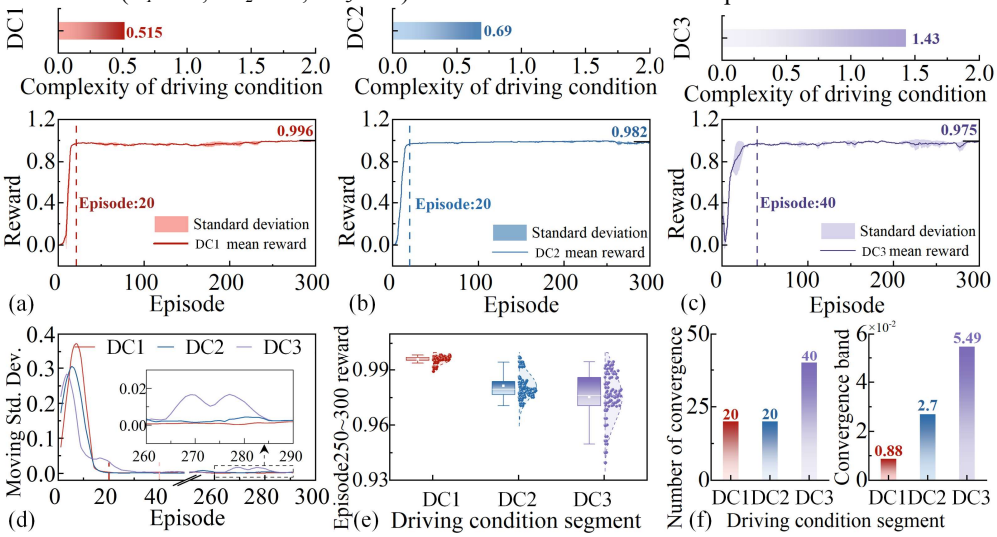


Fig. 7. Training convergence: (a–c) complexity and reward curves for DC1, DC2, and DC3 with increasing complexity; (d) moving standard deviation; (e) reward distribution at episodes 250–300; (f) convergence episodes and bandwidth.

C. Drive-Brake Coordinated Control Framework

Based on the above constraints and problem formulation, a DRL-based coordinated energy management framework for longitudinal drive–brake torque allocation is developed (Fig. 6). Unlike conventional approaches that optimize driving and braking separately, this framework treats positive and negative axle torques as unified control variables and learns a single policy for coordinated torque allocation. This ensures that propulsion, regenerative braking, and mechanical braking are jointly optimized toward the same energy management objective.

As shown in Fig. 6, the framework uses real-time vehicle speed, road slope, pedal commands from driving conditions, and 4WID-EMB feedback as state inputs to ensure energy demand and actuator constraints observability. Based on the constraints of the motor, battery, and braking system, the LSTM-TD3 policy network is trained under the guidance of factors such as power, output torque smoothness, and energy efficiency. The network makes optimal decisions for the front and rear axle torque distribution based on the current input state, outputting torque that balances power, energy efficiency, and smoothness. During online execution, for the left–right axle torque allocation, since lateral control is not considered, the torque is evenly distributed between the left and right wheels on each axle. Additionally, when regenerative braking torque is insufficient, the EMB system supplements the required mechanical braking, achieving coordinated drive–brake control. It should be noted that the torque outputs of the proposed strategy serve solely as energy-optimal reference commands; safety-critical braking functions such as ABS, EBD, and ESP retain full authority and override these references whenever necessary.

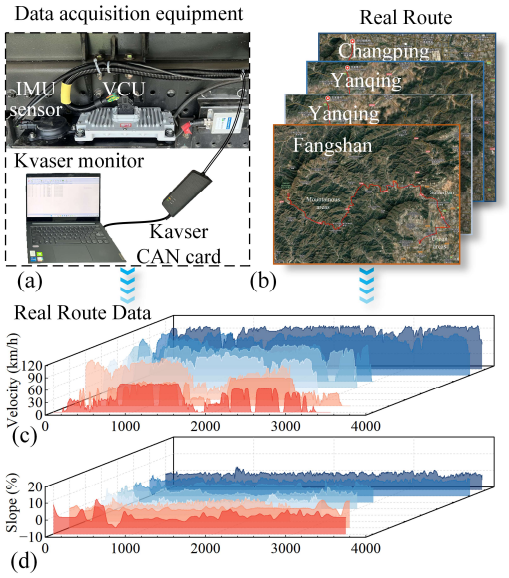


Fig. 8. Data collection process: (a) Data collection; (b) Real Road routes; (c) Velocity data; (d) Slope data.

Therefore, it can be formulated for distributed electric drive configurations that allow independent regulation of front- and rear-axle torques, including but not limited to the 4WID-EMB configuration considered in this work.

IV. Coordination Strategy Training Framework

To learn a nonlinear mapping from high-dimensional vehicle–road states to coordinated drive–brake torque outputs,

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an LSTM-TD3–based policy training framework is developed. By embedding LSTM into TD3, temporal memory is introduced to capture long-term dependencies under complex driving conditions. The influence of the LSTM sequence horizon on training convergence and stability is analyzed to identify an appropriate temporal horizon.

A. TD3 Adaptability Across Driving Conditions

The TD3 algorithm, proven effective in intelligent control, is adopted for strategy. To evaluate its adaptability under varying driving conditions, three representative cycles are selected for training, with driving-condition complexity assessed using velocity, slope, and cycle length as defined in Eq. (14).

$$\lambda_{DCj} = std[N(V_{DCj})] + std[N(i_{DCj})] + N(l_{DCj}), j \in [1,2,3] \quad (14)$$

Where λ_{DCj} is the driving condition complexity, N is the normalization process, and l_{DCj} is the length of the driving condition.

Owing to different cycle lengths, rewards are normalized for comparison. Fig. 7 (a) to (c) respectively illustrate the complexity of three driving conditions and the convergence effects of strategy training. As the complexity of driving conditions increases, the convergence episode gradually expands, and the convergence reward consequently decreases. Training stability is evaluated using moving standard deviation, as shown in Figure (d). For higher-complexity scenarios, the moving standard deviation decreases more slowly and exhibits more pronounced oscillations, indicating slower convergence and poorer stability. Figures (e) and (f) respectively show the reward distribution range, convergence episode count, and convergence reward oscillation bandwidth during the late training phase. These further illustrate that higher driving condition complexity leads to more dispersed reward distributions and poorer convergence stability. Consequently, strategy training for complex long-sequence operational conditions based solely on instantaneous states has limitations.

B. Training Data Source

To validate the adaptability of the proposed energy management strategy under real and complex driving conditions, real vehicle–road driving data were collected for training and validation, as shown in Fig. 8. The test vehicle was equipped with an onboard inertial measurement unit (IMU) operating at a sampling frequency of 100 Hz. Field tests were conducted in several representative traffic regions in Beijing, including Changping, Yanqing, and Fangshan districts, covering various road types such as urban, suburban, highways, and mountainous. The collected dataset captures diverse velocity–slope coupling characteristics. During the data preprocessing stage, the segmentation and classification of driving condition data were performed using the method proposed in our previous work [3].

To ensure the independence between training, validation, and testing, the data segments of each driving-condition category were divided into training, validation, and test sets at a ratio of 7:2:1, and the validation set was used to construct the multimodal validation driving cycle. These segments were combined to form a comprehensive multimodal validation scenario incorporating urban, highway, and grade-driving characteristics, as shown in Fig. 12, and were not used during training. The test set was used for the independent segment-level robustness analysis presented in Section V (Fig. 17). All strategies were trained exclusively on the training segments, and neither the validation nor the test segments participated in training.

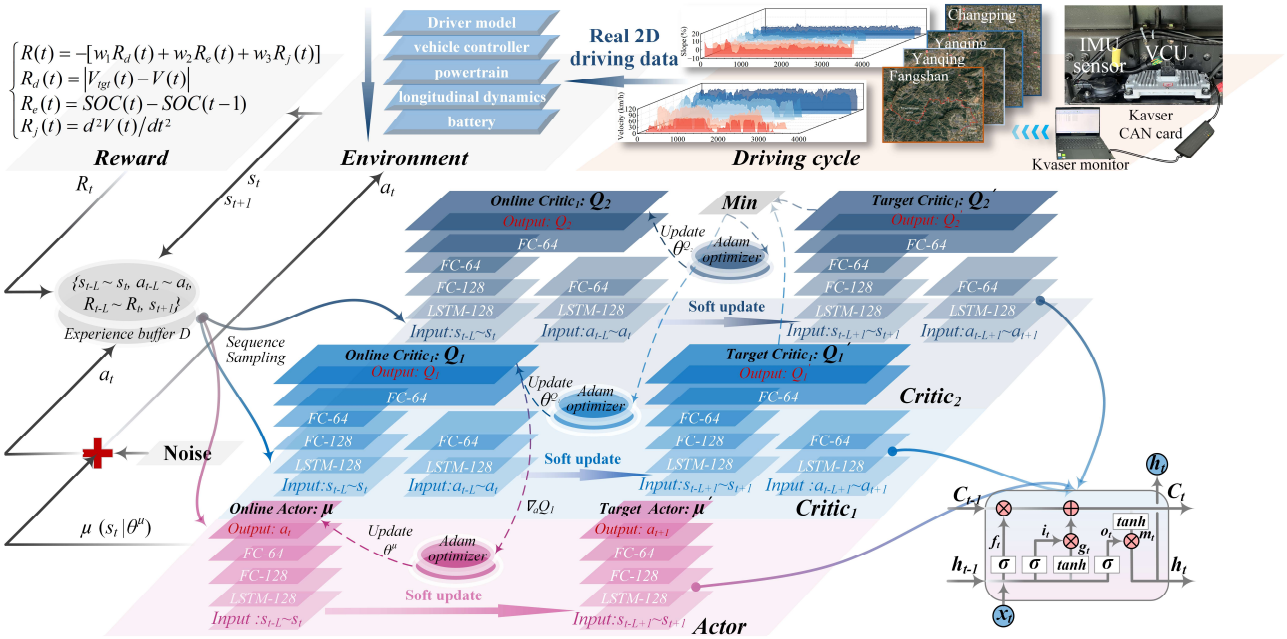


Fig. 9. Training framework for energy-efficient drive-brake coordination control strategy.

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TABLE II
NETWORK ARCHITECTURES AND HYPER PARAMETERS

Parameter description	Value
State dimension	12
State normalization range	[-1, 1]
Action dimension	2
Action range	[-360, 360]
Neurons distribution of LSTM network	128
Input sequence horizon of LSTM (step length)	150
Number of hidden layers of the actor (target) network	2
Neuron distribution of the actor (target) network	128,64
Number of hidden layers of the critic (target) network	3
Neurons distribution of critic (target) network	128,64,64
Optimizer	Adam
Learning rate of the critic	5e-3
Learning rate of the actor	1e-4
Size of experience replay buffer	1e6
Minibatch size	256
Maximum training episodes	300
Discount factor	0.99
Target smooth factor	0.015
Policy update frequency	5
Exploration noise type	Gaussian
Exploration noise standard deviation	0.1
Training sample time	10ms
Random seed	0, 26, 66

C. DRL Framework for Drive–Brake Coordination Strategy

Based on the above analysis, a memory-enhanced LSTM-TD3 training framework for energy-efficient drive–brake coordination is developed (Fig. 9). The framework integrates the environmental model, driving loop, and the defined state/action spaces and reward, while embedding LSTM into TD3 to enable temporal feature extraction.

1) Formulation of LSTM-TD3 Algorithm

The proposed LSTM-TD3 algorithm adopts an Actor–Critic architecture with twin Q-networks to mitigate overestimation bias and improve training stability. As shown in Fig. 9, the framework includes one online Actor, two online Critic networks, and their corresponding target networks. To capture temporal dependencies in driving conditions, LSTM layers are embedded at the input stage of both Actor and Critic networks to process historical state sequences. The extracted temporal features are further mapped through fully connected layers with ReLU activation. The Actor output layer adopts a tanh activation to normalize continuous actions, while the Critic output layer uses linear activation for Q-value estimation. Detailed hyperparameters are summarized in TABLE II.

The policy μ' is trained by maximizing the action-value function in Eq. (15), which represents the cumulative expected reward and thus optimizes long-term energy efficiency.

$$\mu' = arg\ \max_{\mu} \mathbb{E}_{s_t \sim \rho^{\mu}}[Q(S_t, \mu(S_t))] \tag{15}$$

Where μ' is the optimal target strategy, $\mathbb{E}_{s_t \sim \rho^{\mu}}$ represents the expectation under the entire trajectory sampling distribution under the current strategy μ control.

An LSTM network captures long-term temporal dependencies by updating the memory state through selective forgetting and information injection, as described in Eq. (16).

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \tag{16}$$

TD3 employs two critic networks and updates the target Q-value using the minimum of their estimates to mitigate overestimation bias, as given in Eq. (17):

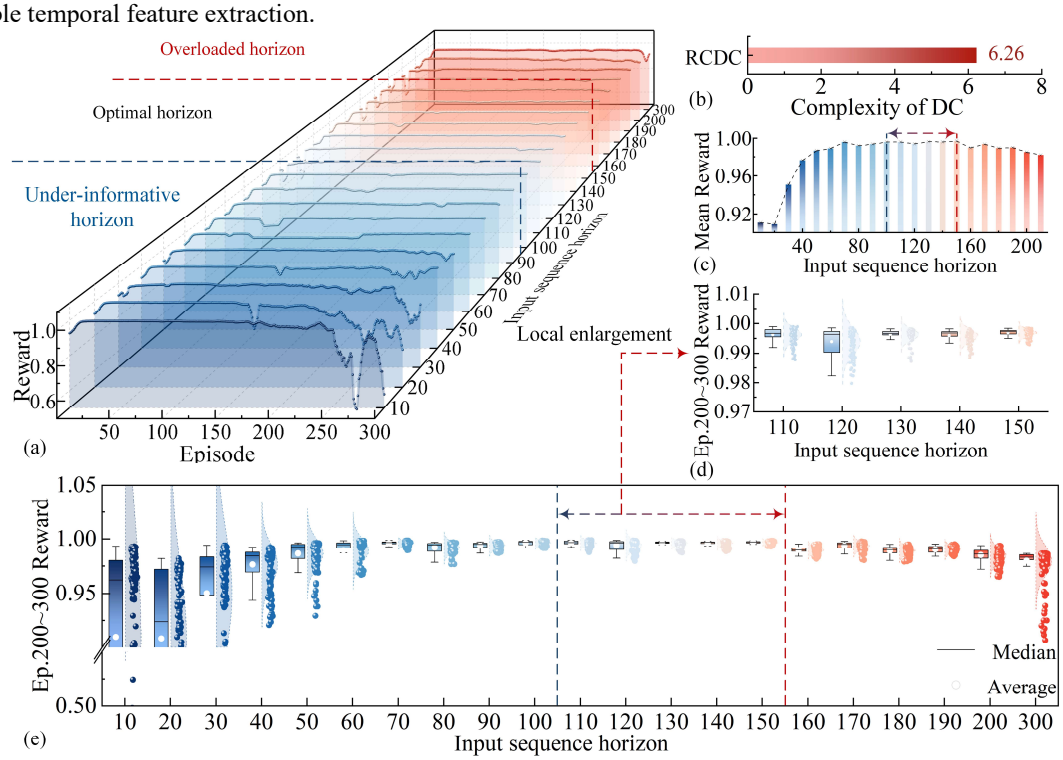


Fig. 10. Effect of LSTM input sequence horizon on training performance: (a) Reward convergence; (b) Driving-condition complexity; (c) Average reward (200–300 episodes); (d) Local zoom of (e); (e) Reward distribution (200–300 episodes).

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$$y_t = R_t + \gamma \min_{i=1,2} Q_i'(S_{t+1}, \mu'(h_{t+1}, S_{t+1})) \quad (17)$$

To enhance temporal modeling, the LSTM module is embedded before the input layers of both actor and critic networks. The actor is optimized by maximizing the critic-evaluated value through the deterministic policy gradient:

$$\nabla_{\theta^\mu} J = \mathbb{E}_{s_t \sim \rho^\mu}[\nabla_a Q(S_t, a)|_{a=\mu(S_t)} \cdot \nabla_{\theta^\mu} \mu(S_t)] \quad (18)$$

With delayed policy updates and target smoothing, the resulting LSTM-TD3 framework achieves stable convergence and improved long-horizon decision quality.

The implementation details of the training process are further specified as follows. Each training episode corresponds to one driving-condition segment randomly sampled from the training set and terminates when the segment ends. Training runs for at most 300 episodes; convergence is identified when the relative variation of the moving-average reward over 10 consecutive episodes falls below 1%, and the converged policy is used for all subsequent evaluations. Both lookup maps, i.e., the motor efficiency map and the battery charge-rate map, are evaluated by linear interpolation. The environment and the controller share a 10 ms time step. Before execution, the raw action is projected onto the feasible torque region defined by the motor external characteristic, the battery charging-power limit, the front–rear braking-distribution boundaries, and the EMB limits; the same projection logic is applied to all compared methods. The overall workflow is summarized in Algorithm 1.

Algorithm 1 Training and Evaluation Procedure of the Proposed Memory-Enhanced TD3 Framework

Input: Driving dataset D, vehicle parameters, constraint set C, and training hyperparameters.
Output: Trained memory-enhanced front-rear torque-allocation policy π_θ.

// Phase 1. Data preparation
Segment and classify the collected driving data into DC1-DC6;
Split each category into training, validation, and test sets (7:2:1);
Construct the multimodal validation cycle from the validation set;
// Phase 2. Environment and network initialization
Construct the longitudinal vehicle dynamics, powertrain, and battery models;
Define motor, battery, braking-distribution, and EMB constraints;
Initialize Actor π_θ, twin Critics $Q_{\varphi 1}$ and $Q_{\varphi 2}$, and target networks;
Initialize replay buffer B;
// Phase 3. Memory-enhanced TD3 training
for episode = 1 to 300 **do**
  Randomly sample a driving segment from the training set;
  Reset the environment and initialize the state-sequence buffer;
  for each 10-ms control step t **do**
    Construct the Th-stepstate sequence $s[t-T_h+1:t]$ (T_h = 150);
    Normalize the state sequence to [-1, 1] using min-max normalization;
    Select $a(t) = \pi_\theta(s[t-Th+1:t]) + \varepsilon$, $\varepsilon \sim N(0, \sigma^2)$, $\sigma = 0.1$;
    Project a(t) onto the feasible front-rear torque region;
    Execute a(t) and update the vehicle environment and SOC;
    Compute r(t) from the normalized tracking, energy-consumption, and jerk terms;
    Store the transition in replay buffer B;
    Sample a minibatch of 256 transitions from B;
    Compute the smoothed target action;
    Update $Q_{\varphi 1}$ and $Q_{\varphi 2}$ using the minimum target Q-value;
    if t mod 5 = 0 **then**
      Update Actor π_θ;
      Soft-update the target networks with $\tau = 0.015$;
    end if
  end for
end for
// Phase 4. Policy evaluation
Repeat the training procedure using three independent seeds;
Report the training statistics as mean ± standard deviation;
Evaluate the converged policies on the validation cycle;
Validate real-time execution on the HIL platform;
Evaluate the three independently trained policies on the held-out test segments;
return π_θ;

2) **Input Sequence Horizon of LSTM-TD3 Algorithm**

Based on the proposed architecture, multiple training experiments reveal that although rewards converge smoothly in early stages, mid-to-late training may exhibit sudden drops or oscillations (e.g., T_h=10 in Fig. 10(a)). This instability is attributed to catastrophic forgetting in LSTM-based DRL, where shifts in state–action distributions cause drift in temporal representations and abrupt policy changes. Although early stopping can suppress late-stage oscillations, it limits exposure to complex and delayed-feedback scenarios, leading to suboptimal generalization and reduced long-term robustness. We further observed that the LSTM input sequence horizon has a pronounced influence on training stability. Varying the horizon alters the policy's temporal modeling granularity and its ability to capture state-transition patterns, thereby affecting convergence performance. To quantify this effect, we conducted policy training under different sequence horizons from T_h=10 to 200, (step size = 10ms) and additionally evaluated T_h=300 within the constraint of acceptable training time. The results are shown in Fig. 10.

Fig. 10(a) illustrates reward convergence under different input sequence horizons. Short horizons lead to late-stage oscillations or divergence due to insufficient temporal information, while increasing the horizon improves training stability. However, excessively long horizons (e.g., T_h=300) may also reintroduce instability. When the temporal context becomes overly long, the policy must infer the impact of current actions on rewards from a much longer history, which increases the difficulty of temporal credit assignment. Meanwhile, redundant historical information may dilute critical temporal features in the LSTM memory state, reducing the effectiveness of temporal abstraction. These unstable temporal representations can further propagate estimation errors in the critic network, resulting in oscillatory policy updates and reward. Accordingly, the temporal horizon can be divided into three characteristic regions:

- Under-informative horizon: Historical information is inadequate for the LSTM to extract temporal dependencies.
- Optimal horizon: the provided temporal horizon context captures essential causal patterns for policy generation, achieving stable training without redundant inputs.
- Overloaded horizon: excessive historical information introduces non-causal redundancy, destabilizes memory representation, and aggravates catastrophic forgetting.

Fig. 10 (b) illustrates the complexity of the test conditions. Fig. 10 (c) reports the average reward between episodes 200-300, where the optimal input-sequence horizon achieves the

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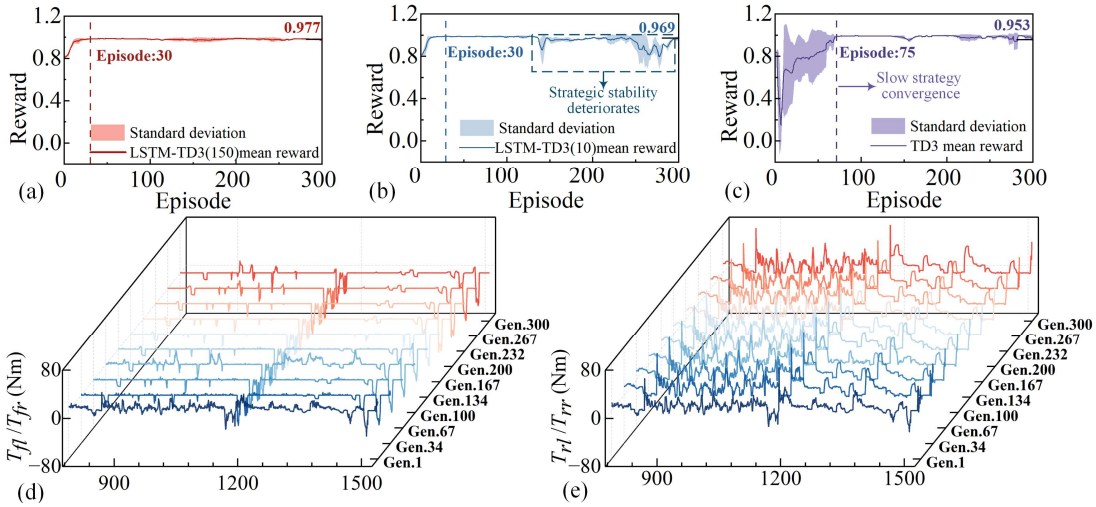


Fig. 11. Comparison of the training convergence effects of strategies: (a) LSTM-TD3 (T_h=150) reward; (b) LSTM-TD3 (T_h=10) reward; (c) TD3 reward; (d) Variation curve of front axle torque; (e) Variation curve of rear axle torque.

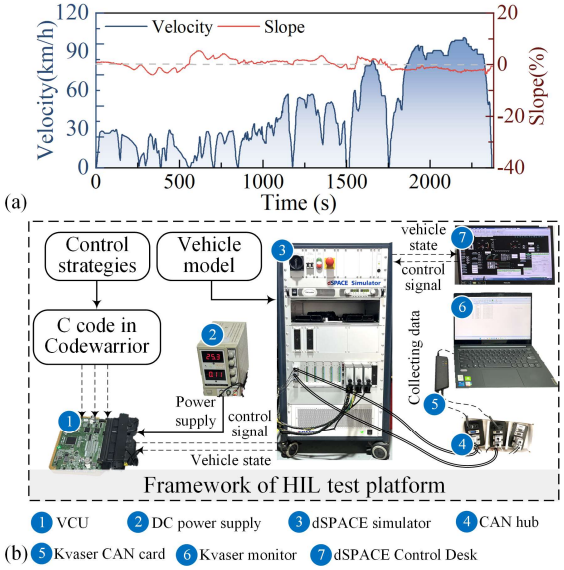


Fig. 12. Validation driving cycle and HIL test platform.

highest converged reward, indicating superior energy-management performance. Fig. 10 (e) presents the reward distribution in the same interval. The optimal horizon exhibits a compact and low-variance distribution, reflecting stable training behavior, whereas horizons farther from this range display increasingly dispersed rewards and pronounced oscillations. This confirms the effectiveness of the optimal temporal horizon in enhancing training robustness and control consistency. Finally, two aspects of the horizon selection are further clarified. First, the optimal horizon was identified from training-side convergence behavior, namely a high converged reward accompanied by stable late-stage training without divergence, rather than from validation-cycle energy or smoothness metrics. Second, with the 10 ms control step, the selected horizon Th=150 corresponds to a temporal context of 1.5 s, which spans the dominant time scale of the longitudinal drive–brake response, including torque buildup and braking transients. Horizons far below this scale cannot fully capture the transient behavior, whereas horizons much longer than this scale extend beyond the physically relevant response window and introduce redundant historical information, which is consistent with the three-region division discussed above.

V. Results And Discussion

This section evaluates the convergence and performance of the proposed LSTM-TD3 coordinated front–rear torque allocation framework.

A. Convergence Effect of Strategy Training

A coordinated energy management policy trained with the proposed LSTM-TD3 framework is compared with TD3 without memory and LSTM-TD3 using an unoptimized sequence horizon. All algorithms are trained using three independent random seeds, and the reward curves represent the mean values across runs, while the shaded regions indicate the standard deviation.

Fig. 11(a)~(c) respectively show the training convergence of the LSTM-TD3 (T_h=150), LSTM-TD3 (T_h=10), and TD3 algorithms under complex operating conditions. TD3 converges slowly and exhibits instability under long-term energy-demand dependencies. While LSTM improves temporal feature extraction, an unoptimized horizon causes mid- and late-stage reward oscillations and even policy collapse (Fig. 11(b)), indicating that sequence horizon critically affects training stability and generalization. In contrast, the optimized-horizon LSTM-TD3 achieves fast, stable convergence and robust performance, demonstrating superior capability in capturing long-term energy-related temporal correlations. Training convergence is identified when the moving-average reward stabilizes, and the variation over consecutive episodes becomes negligible.

Fig. 11(d) and (e) illustrate the evolution of front–rear torque allocation during training. In the early stage, torque distribution is highly random and unstable, indicating insufficient learning of energy-efficient allocation. After convergence, a clear pattern emerges: rear-axle–preferred

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torque during driving and front-axle–preferred torque during regenerative braking, reflecting an energy-optimal strategy consistent with vehicle dynamics constraints. Rear-axle driving reduces operation in inefficient motor regions, while front-axle–dominant braking exploits load transfer to enhance regenerative capability. Meanwhile, torque outputs become smoother, reducing torque switching and longitudinal jerk, improving ride comfort and energy management stability.

Furthermore, the sensitivity of the strategy performance to the reward weights was examined through a perturbation experiment, as summarized in TABLE III. Starting from the selected combination (w_1=0.3, w_2=0.6, w_3=0.1), each weight was locally perturbed while the remaining training settings, test conditions, and constraint-handling logic were kept unchanged. Increasing the dynamic-performance weight reduces the velocity-tracking error but increases energy consumption and jerk; increasing the energy weight further reduces EC/100 km at the cost of tracking accuracy and smoothness; and increasing the smoothness weight makes the strategy more conservative. The selected combination is not optimal for any single metric but achieves the most balanced overall performance, which is consistent with the economy-oriented reward design described in Section III-B.

TABLE III
EFFECTS OF REWARD WEIGHTS ON STRATEGY PERFORMANCE

Case	w_1	w_2	w_3	EC/100 km (kWh)	Velocity RMSE (km/h)	RMS jerk (m/s3)
Base	0.3	0.6	0.1	11.56	0.989	0.391
Dynamic-low	0.2	0.65	0.15	11.51	1.25	0.388
Dynamic-high	0.4	0.55	0.05	11.82	0.901	0.471
Economic-low	0.35	0.5	0.15	12.19	0.952	0.389
Economic-high	0.25	0.7	0.05	11.48	1.173	0.473
Smoothness-low	0.35	0.65	0	11.53	0.949	0.542
Smoothness-high	0.25	0.55	0.2	11.81	1.171	0.386

B. Performance of Driving–Braking Coordination EMS

The trained coordinated strategy is integrated into the energy management framework and evaluated under complex driving conditions. As shown in Fig. 12 (a), the validation is conducted using a real vehicle–road multimodal cycle constructed from measured driving data, including representative urban, highway, and grade-related operating segments. The segment extraction principles follow our previous work [3]. To further improve reproducibility, the detailed composition ratios of different driving segments, the raw validation-cycle data are provided in the public GitHub repository established for this work. The constructed validation cycle is then extended to a dSPACE-based hardware-in-the-loop (HIL) platform to examine the real-time execution capability and hardware compatibility of the proposed strategy.

As shown in Fig. 12 (b), the HIL platform mainly consists of a dSPACE Modular HIL Simulator equipped with a DS1006 Processor Board (quad-core AMD Opteron processor, 2.8 GHz), a vehicle control unit (VCU), the dSPACE ControlDesk experiment software, a CAN communication interface, a signal acquisition module, and a power supply module. In the HIL platform, the distributed electric vehicle model, built in Simulink and compiled into a real-time application via Simulink Coder with the dSPACE Real-Time Interface, is executed on the dSPACE real-time simulator, while the proposed Actor policy network is converted into C code and deployed on a production automotive-grade VCU for closed-loop validation. The simulator and controller are both configured with a 10 ms sampling interval. During real-time operation, vehicle-state signals, including vehicle velocity, road slope, pedal information, SOC, and other relevant states, are transmitted from the dSPACE simulator to the VCU through the CAN interface, while the front and rear axle torque commands generated by the VCU are sent back to the simulator for execution. Moreover, to ensure the safety and physical implementability of the torque allocation command, the command generated by the VCU is checked against the motor torque limits, front-rear torque distribution constraints, and battery regenerative braking power constraints before execution. If any constraint is violated, the command is corrected before being applied. The controller execution and HIL validation are carried out under the component-level operating limits listed in TABLE I, thereby ensuring that the real-time control commands remain feasible within the considered model constraints and compatible with the assumed hardware configuration.

TABLE IV
EXECUTION-TIME BUDGET OF ONE CONTROL STEP WITHIN THE 10 MS CONTROL CYCLE

Component	Time (ms)	Share of the 10 ms cycle
Signal reception and preprocessing	≈0.5	≈5%
Actor forward inference	0.206	2.06%
Constraint checking and projection	<0.05	<0.5%
CAN transmission of torque commands	≈0.25	≈2.5%
Total closed-loop execution time	<2	<20%

In addition, the computational overhead and execution timing of the deployed strategy were evaluated. During online operation, only the pretrained Actor network is executed on the VCU, while the Critic networks do not participate in online inference. The Actor is converted into C code and deployed in single-precision floating-point (FP32) format, with 96,578 parameters occupying approximately 377.3 KB; owing to the compact network size, no pruning or quantization is required. The deployed control process occupies 16.22 MB of runtime memory, and its CPU utilization remains at approximately 0.3% on average with an instantaneous peak of 0.76%. The execution time of one control step consists of signal reception and preprocessing, Actor inference, constraint checking and projection, and CAN transmission of the torque commands at 500 kbps; a single forward inference takes 0.206 ms (2.06% of the 10 ms control cycle), and the total closed-loop execution time remains below 2 ms, i.e., within 20% of the control cycle (TABLE IV), leaving a timing margin of more than 80%. Therefore, the proposed approach exhibits low computational overhead and satisfies the real-time

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execution requirements of online energy management under the present HIL-based implementation setting.

1) **Velocity Tracking Performance**

In this study, velocity-tracking performance is not treated as the primary optimization objective, since sufficient optimization freedom should be reserved for energy saving. Instead, it is used as a feasibility criterion to ensure that the basic driving demand is satisfied. Therefore, the velocity-tracking result is presented only to verify whether the proposed strategy satisfies the dynamic constraints [44]. As shown in Fig. 13, V_{real} follows V_{tgt} closely, with the peak error remaining below 4.2 km/h, and the RMSE is less than 1 km/h.

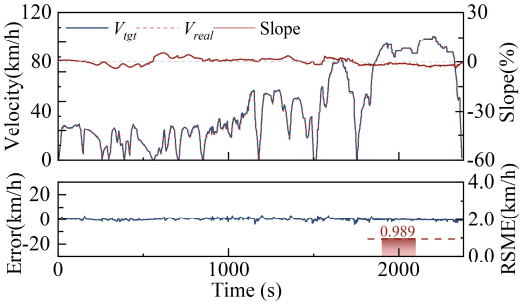


Fig. 13. The vehicle's velocity tracking performance.

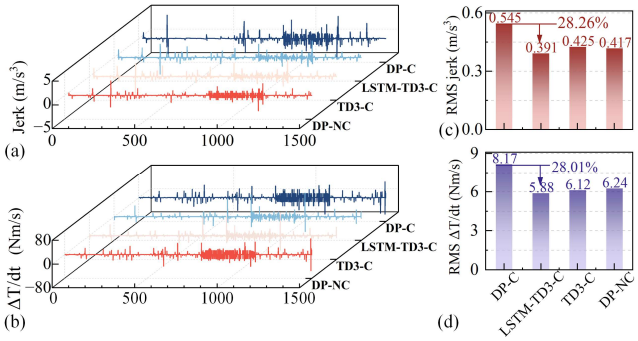


Fig. 14. Smoothness comparison: (a) longitudinal jerk; (b) rear-axle torque derivative; (c) RMS jerk; (d) RMS rear-axle torque derivative (DP-C: DP-coordinated, TD3-C: TD3-coordinated, DP-NC: DP-non-coordinated, LSTM-TD3-C: LSTM-TD3-coordinated)

These results demonstrate that the proposed strategy can satisfy the target driving demand and basic tracking requirements under the considered validation cycle [45].

2) **The Smoothness of Torque Distribution**

To evaluate torque-distribution smoothness, longitudinal jerk and torque variation rate ($\Delta T/dt$) are employed as engineering indicators of control-output smoothness.

As shown in Fig. 14 (a)–(b), the DP-based methods exhibit significant jerk spikes and torque-rate oscillations. The phenomenon is particularly obvious in DP-C, where the discrete optimization process over a broader feasible domain tends to introduce larger inter-step torque variations. Although DP-NC is relatively more stable, it still presents visible oscillatory behavior. By contrast, LSTM-TD3-C maintains a smoother torque evolution throughout the cycle. This result suggests that the temporal-memory mechanism helps the controller account for historical operating information, thereby improving continuity in torque regulation under complex long-horizon driving conditions.

The cycle-level RMS results in Fig. 14(c)–(d) further show that LSTM-TD3-C achieves lower jerk and ($\Delta T/dt$), indicating improved fluctuation suppression and smoother torque output under the considered validation conditions.

3) **Energy Efficiency Performance**

After verifying compliance with dynamic and smoothness constraints, the remaining optimization capacity is devoted to energy efficiency. Accordingly, energy efficiency is assigned the highest reward weight via SOC variation to maximize energy savings under the prescribed constraints.

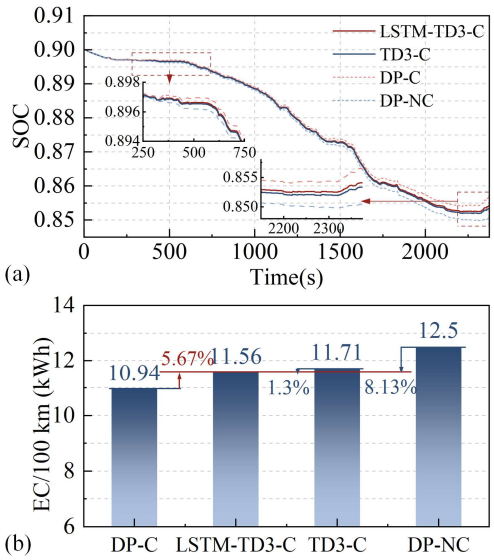


Fig. 15. Comparison of SOC and EC/100 km under different control strategies: (a) SOC; (b) EC/100km.

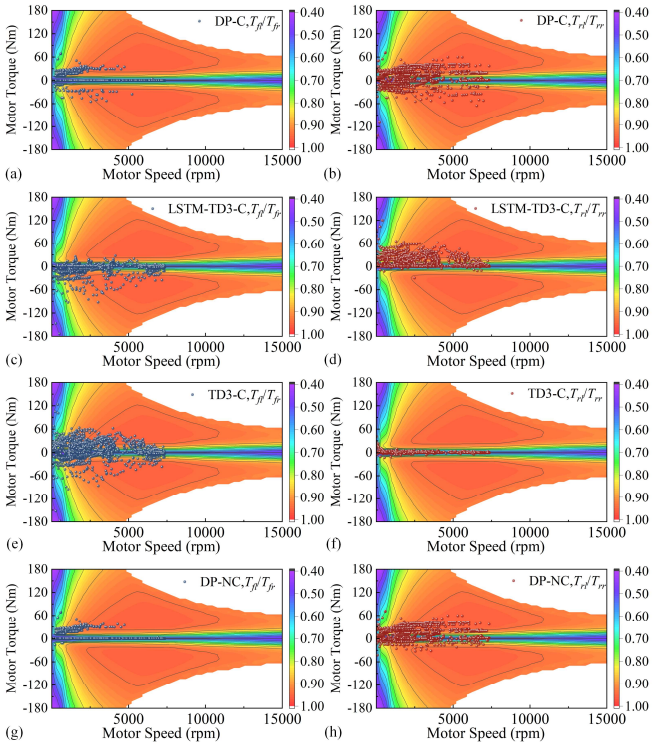


Fig. 16. Comparison of motor operating-point distributions: (a–b) DP-C; (c–d) LSTM-TD3-C; (e–f) TD3-C; (g–h) DP-NC.

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Fig. 15 compares the SOC curves and 100-kilometre energy consumption performance of LSTM-TD3-C with those of DP-C, DP-NC, and TD3-C. It should be emphasized that DP-C and DP-NC are dynamic programming (DP)-based offline references [46] rather than online-deployable strategies, since DP requires complete prior knowledge of the entire driving cycle. Both DP references are solved subject to the same physical powertrain constraints as the online strategies, including the motor torque–speed limits, the battery charging-power limits, and the actuator feasibility boundaries listed in TABLE I. However, the braking-stability-related front–rear braking-force distribution constraints defined by the I-curve and the ECE regulation in Section III-A are not imposed in DP-C. DP-C therefore characterizes a relaxed theoretical lower bound on energy consumption for the unified drive–brake coordination framework, rather than the offline optimum of the same constrained problem solved by the proposed online controller. DP-NC denotes the corresponding offline reference under the non-coordinated framework, in which the braking demand is preferentially supplied by mechanical braking.

The results show that, compared to the non-coordination offline reference (DP-NC), all other coordination strategies exhibit lower energy consumption levels. This indicates that the introduction of a drive-brake cooperation strategy enables more efficient utilization of regenerative braking capabilities, thereby effectively reducing the vehicle's energy consumption. Based on this, the proposed online learning method is further compared with DP-C. Since DP-C solves a relaxed problem, its energy consumption provides a lower bound for any controller that enforces the complete braking-stability constraint set, making this comparison conservative. Under the present online HIL-based implementation conditions, the proposed LSTM-TD3-C explicitly enforces the braking-stability-related torque-distribution constraints online while achieving energy consumption only 5.67% above this relaxed theoretical lower bound; its deviation from the true optimum of the constrained problem is therefore even smaller. In contrast, the TD3-C exhibits a larger deviation from the DP-C lower bound, and its energy consumption is 1.3% higher than that of the LSTM-TD3-C. This indicates that, following the introduction of temporal feature modeling, the proposed method achieves more stable and efficient energy allocation under complex driving conditions. Therefore, while satisfying the real-time execution constraints of the present HIL-based implementation, the proposed method closely approaches the practically attainable energy limit for drive-brake coordination.

To further interpret the observed energy-consumption differences, Fig. 16 analyzes the distribution of motor operating points. As shown in (a)-(b), DP-C concentrates operation within the high-efficiency region, which is consistent with its role as a relaxed energy lower bound. Notably, DP-C relies predominantly on the rear motor for both driving and regenerative braking. This behavior is admissible in the relaxed problem precisely because the braking-stability-related front–rear distribution constraints are not imposed, but it would not be feasible for a stability-constrained controller during real braking events. The operating-point distribution of DP-C thus reflects the difference in feasible sets rather than a deployable allocation pattern.

In contrast, the proposed LSTM-TD3-C strategy exhibits a rear-axle preference during driving and a front-axle preference during braking, as shown in (c)-(d). This pattern increases the likelihood of motor operation within high-efficiency zones while simultaneously satisfying braking-stability requirements. For TD3-C, limited temporal-dependency modeling leads to unstable long-horizon behaviors. As shown in (e)-(f), although TD3-C increases front-axle torque during braking to improve stability, it struggles to keep both motors operating efficiently over extended sequences. (g)-(h) further show that DP-NC rarely utilizes negative-torque operating regions, indicating that mechanical braking dominates and much of the recoverable energy is lost as frictional heat.

Overall, SOC trends, EC/100 km, and operating-point distributions confirm effective front–rear power allocation, enhanced use of regenerative braking, and improved energy efficiency under the considered validation conditions.

To verify that the energy-saving advantage of the proposed method does not depend on a specific training run, the two learning-based strategies were trained with three independent random seeds (0, 26, and 66), and each converged policy was evaluated on the same validation cycle, as summarized in TABLE V. The proposed LSTM-TD3-C achieves a lower mean EC/100 km than TD3-C (11.56 vs. 11.71 kWh) with a smaller standard deviation (0.02 vs. 0.061 kWh), indicating that its performance improvement is consistent across different training initializations.

TABLE V
ENERGY CONSUMPTION OF THE LEARNING-BASED STRATEGIES UNDER DIFFERENT RANDOM SEEDS

Method	Random seeds	EC/100 km of each seed (kWh)	Mean ± Std (kWh)
TD3-C	0, 26, 66	11.78 / 11.68 / 11.67	11.71 ± 0.061
LSTM-TD3-C	0, 26, 66	11.58 / 11.56 / 11.54	11.56 ± 0.02

To verify whether the energy-saving performance of the proposed method depends on specific validation scenarios, this work extracted 30 independent validation segments (lengths ranging from 2.1 to 5.5 km) from real-world vehicle-road multimodal data that had not been used in training. For each method, every segment was evaluated using the three independently trained policies (random seeds 0, 26, and 66), and the segment-level EC/100 km was averaged over the three runs.

As shown in Fig. 17(a), due to differences among the driving segments in terms of segment length, velocity variations, slope changes, and start-stop characteristics, the EC/100km exhibits some variability across different segments. Nevertheless, the energy consumption distribution of the proposed LSTM-TD3-C method is generally lower than that of TD3-C and DP-NC, indicating that its energy-saving advantage is not limited to any specific validation condition.

Considering the varying driving distances across different driving conditions, this work employs a distance-weighted approach to calculate the overall EC/100km, in which the EC/100 km of each segment is weighted by its travelled

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distance and the summed value is normalized by the total distance, so that short segments are not overrepresented. As shown in Fig. 17(b), LSTM-TD3-C achieves a reduction of 2.35% and 8.57% compared to TD3-C and DP-NC, respectively. It also remains the closest to the DP-C theoretical lower bound among the online strategies, further demonstrating that the proposed method maintains stable energy-saving performance across diverse validation driving conditions.

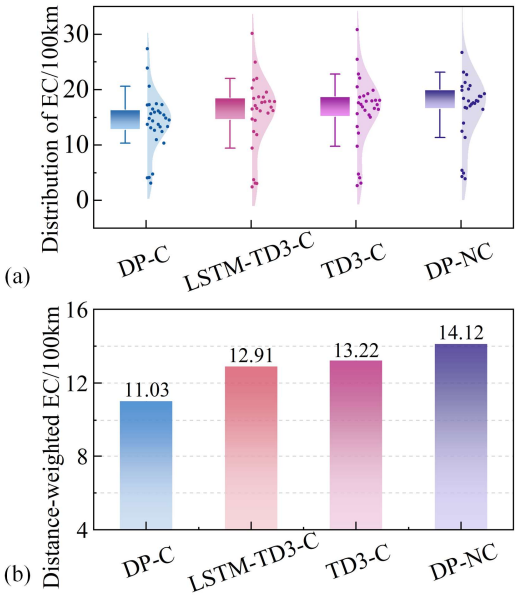


Fig. 17. Statistical robustness analysis of EC/100 km across 30 independent validation driving segments: (a) segment-level EC/100 km distribution; (b) distance-weighted average EC/100 km.

VI. Conclusion

This research proposes a drive-brake coordinated energy management framework for distributed electric vehicles capable of independent front–rear torque regulation, unifying positive and negative wheel torques into a single control variable to achieve unified optimization of the drive and regenerative braking processes. To address the challenges of online decision-making under conditions of strong actuator coupling and long-duration driving sequences, a LSTM-TD3 control strategy training architecture with temporal memory capabilities was developed. By analyzing the input sequence length, an effective sequence range was identified that supports stable strategy convergence and long-term energy optimization. The results of the research indicate that:

- **The drive-brake coordination mechanism can significantly improve energy utilization efficiency.** Compared to the non-coordinated offline reference benchmark DP-NC, the drive-brake coordination strategy effectively reduces energy waste caused by mechanical braking priority and enhances regenerative braking energy recovery capabilities. Notably, the proposed method achieves an 8.13% reduction in Energy consumption (EC) /100km compared to DP-NC, indicating the energy-saving potential of the drive-brake coordination control architecture under the considered validation conditions. In addition, the energy-saving advantage remains consistent across diverse validation driving segments, further demonstrating the statistical robustness of the proposed method.
- **The introduction of temporal memory further enhances the convergence, stability, and generalization capabilities of the reinforcement learning strategy.** Compared to the memory-less TD3-C, LSTM-TD3-C is better able to capture state dependencies in long-term torque distribution, thereby achieving more stable strategy outputs under complex vehicle–road coupling conditions. Its EC/100km is reduced by an additional 1.3% compared to TD3-C.
- **The proposed strategy demonstrates real-time execution feasibility in the present HIL-based implementation while achieving a closer approximation to the offline theoretical energy lower bound.** The proposed method satisfies longitudinal dynamic constraints and maintains a smooth torque output. Furthermore, its energy consumption is only 5.67% above the relaxed theoretical lower bound provided by DP-C, indicating that the method closely approaches the practically attainable energy limit while enforcing braking-stability constraints under real-time online decision-making conditions.

Although validated on a four-wheel distributed drive configuration, the proposed framework essentially addresses front–rear torque allocation. Thus, it can be extended to other distributed drive configurations with independent front–rear torque allocation capability, provided that the environment-model parameters and configuration are updated according to the target platform.

Overall, the LSTM-TD3-based framework achieves stable convergence, satisfactory longitudinal performance, and improved energy efficiency under the considered validation and HIL test conditions. Future work will include real-vehicle closed-loop road validation, the incorporation of emergency-braking safety constraints to extend the framework toward broader driving scenarios, and further exploration of reinforcement-learning-based left–right differential torque control to enhance yaw stability.

Appendix A. Supplementary Data

For reproducibility, additional details on the environmental model setup, constraint implementation, training configuration, and the construction of the multimodal validation cycle are provided in the GitHub repository for this study: https://github.com/12138Armin/Distributed-ev-ems

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Huimin Liu received the B.S. degree in Vehicle Engineering from China Agricultural University, Beijing, China, in 2021. She is currently pursuing the Ph.D. degree in Mechanical Engineering with Beijing Institute of Technology, Beijing, China. Her research interests include distributed-drive electric vehicles, vehicle dynamics control, deep reinforcement learning and energy management strategy.



Cheng Lin received the Ph.D. degree in mechanical engineering from Beijing Institute of Technology, Beijing, China, in 2004. He is currently a Professor at the School of Mechanical Engineering, Beijing Institute of Technology, Beijing, China. His research interests include electric vehicle powertrain system platform technology, vehicle electric drive and intelligent control technology, and intelligent power battery technology.



Xiao Yu received the M.S. degree in vehicle engineering from Shandong University of Technology, Zibo, China, in 2020, and the Ph.D. degree in mechanical engineering from Beijing Institute of Technology, Beijing, China, in 2025. He is currently a Lecturer with Shandong University of Technology, Zibo, China. His research interests include configuration design and optimization of electric vehicle drive systems, intelligent vehicle control under complex operating conditions, and multimodal fusion perception and collaboration.



Zhenyi Tao received the B.S. degree in Mechanical Engineering from Beijing Institute of Technology, Beijing, China, in 2021. He is currently pursuing the Ph.D. degree in Mechanical Engineering with Beijing Institute of Technology, Beijing, China. His research interests include electric vehicle dynamics control, intelligent perception of vehicle states and energy management strategy.



Xingyu Xia received the B.S. degree in Vehicle Engineering from Beijing Institute of Technology, Beijing, China, in 2025. He is currently pursuing the Ph.D. degree in Mechanical Engineering with Beijing Institute of Technology, Beijing, China. His research interests include intelligent vehicle state estimation and perception, and vehicle dynamics control for distributed-drive electric vehicles.



Jiaqi Xu received the B.S. degree in Aerospace engineering from Beijing Institute of Technology, Beijing, China, in 2021. He is currently pursuing the Ph.D. degree in Mechanical Engineering with Beijing Institute of Technology, Beijing, China. His research interests include electric drive system, gear shift strategy, and energy manage strategy.



Liqing Zhu received the B.S. degree in Vehicle Engineering from Beijing Institute of Technology, Beijing, China, in 2024. He is currently pursuing the Ph.D. degree in Mechanical Engineering with Beijing Institute of Technology, Beijing, China. His research interests include Model Predictive Control, Data-driven modeling approach, and vehicle dynamics.



Jinhua Wen received the B.S. degree in Vehicle Engineering from China Agricultural University, Beijing, China, in 2024. He is currently pursuing the M.S. degree in Vehicle Engineering at Beijing Institute of Technology, Beijing, China. His research interests include distributed-drive electric vehicles, vehicle dynamics control, deep reinforcement learning.

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